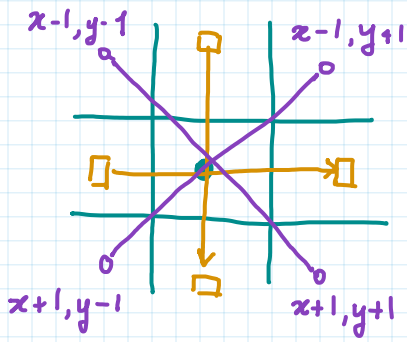

FILTRO di SOBEL

$$M \in \mathbb{R}^{3\times 3} \quad \swarrow \quad \searrow$$

$$M_x \qquad M_y$$

$h = k = -1$

$$f(x-1, y-1) \cong f(x,y) - \frac{\partial f}{\partial x}(x,y) - \frac{\partial f}{\partial y}(x,y)$$

$h = k = 1$

$$f(x+1, y+1) \cong f(x,y) + \frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y)$$

$$\frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}(x,y) \cong \frac{f(x+1,y+1) - f(x-1,y-1)}{2}$$

HW : fare le altre APPROX.